🧪 POLYMER — FULL MIND MAP

Chemistry · Chapter 7.C · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

📄 Topics in this chapter:

1. Polymer Basics & Natural Polymers

- Polymerization mein bahut saare small molecules (monomers) judkar large polymer molecule banate hain. Natural rubber ka monomer Isoprene hai; isliye natural rubber ko polyisoprene maana jata hai. [Q1–Q2]
- Wool, Silk aur Leather natural polymers hain, jabki Nylon synthetic polymer hai. Silk khud ek natural protein polymer hai; Ghee lipids/fats ka mixture hai, polymer nahi. Starch, Protein aur Cotton (cellulose) natural polymers ke examples hain. [Q3–Q5]
- Silk ka major structural protein Fibroin hai. Protein amino acids se bante hain, isliye natural silk mein Nitrogen important constituent element hai. [Q6]
- Cellulose aur Starch dono glucose units se bane polysaccharides hain. Cellulose nature ka sabse abundant organic compound maana jata hai aur carbohydrate group se belong karta hai; cotton cellulose ka bahut pure natural source hai. [Q7–Q9]
- Rayon, linen aur jute largely cellulosic fibres hain; Nylon synthetic aur non-cellulosic fibre ka example hai. [Q10]

2. Fibre / Plastic Classification

- Artificial silk ka naam Rayon se associated hai, Acrylic se nahi. Rayon chemically treated wood pulp/cellulose se banta hai; Acrylic fibres natural wool jaise properties dikhate hain. Polycarbonate CDs mein aur Teflon non-stick coating mein use hota hai. [Q11]
- Neoprene thermoplastic nahi, balki synthetic-rubber elastomer hai. Ye chloroprene ke polymerization se banta hai aur polychloroprene bhi kaha jata hai. [Q12]
- PVC (polyvinyl chloride) common synthetic polymer hai jo readily biodegradable nahi hota; cellulose, starch aur proteins natural biodegradable polymers hain. [Q13]
- Melamine ek thermosetting plastic hai. Heat/cure ke baad iska structure permanently rigid ho jata hai aur ise remelt karke thermoplastic ki tarah reshape nahi kiya ja sakta; floor tiles/laminates/kitchenware mein use hota hai. [Q17]

3. Polythene, PET & Hydrogels

- Polythene/Polyethylene ka actual monomer Ethylene (Ethene, C_2H_4) hai. Ethylene ke addition polymerization se polyethylene banta hai. [Q14–Q15]

⚠️ EXAM TRAP — Q14 mein correct monomer option hi missing hai

▸ Polythene ka monomer Ethylene hai, lekin Q14 ke options Methane/Ethane/Propane/Butane the; isliye source ne ise disputed (*) mark kiya hai.
▸ Ethane aur Ethylene ko confuse mat karna: polymerization ke liye unsaturated Ethylene (C_2H_4) chahiye.

- PET (Polyethylene terephthalate) fibres ko wool/cotton ke saath blend kiya ja sakta hai aur PET bottles recycle hokar carpets, packaging ya clothing jaise products mein ja sakti hain. “Any alcoholic beverage storage” aur “incineration without greenhouse gases” statements correct nahi hain. [Q18]
- Hydrogels water-rich polymeric networks hote hain. Source ke according controlled drug delivery, mobile cooling/air-conditioning applications aur industrial lubrication mein inka use ho sakta hai. [Q19]

⚠️ EXAM TRAP — Q16 ka “polyethylene gas” wording scientifically misleading hai

▸ Polyethylene khud high-molecular-weight polymer hai aur normal conditions mein gas nahi hota.
▸ Q16 ko source-mapping ke liye preserve kiya gaya hai, lekin “plastic se polyethylene gas milti hai” ko standard chemistry fact ke roop mein yaad na karein. [Q16]

4. Bulletproof Polymers & Armour Materials

- Kevlar lightweight bulletproof jackets/vests aur helmets ka classic fibre hai. Ye high tensile-strength para-aromatic Polyamide synthetic fibre hai. [Q20–Q24]
- Bullet-resistant windows/transparent armour mein transparent Polycarbonate (e.g., Lexan) aur laminated-glass layers use hoti hain; Kevlar opaque body armour ke liye more appropriate hai. [Q25]

⚠ EXAM TRAP — Q26 — "Bulletproof jacket" vs Laminated Glass

- Source ke historical official key mein Laminated Glass accept hua, lekin actual bulletproof jacket/body armour ka standard fibre Kevlar hota hai.
- Laminated glass/polycarbonate ko transparent armour/windows se associate karo; Kevlar ko vests/jackets se.

- Bulletproof materials ke context mein Kevlar + Lexan dono relevant hain: Kevlar body armour, Lexan polycarbonate transparent armour/glass. Glyptal paints/lacquers ke protective coatings mein use hota hai. [Q26–Q27]

5. Neoprene, Teflon & Common Synthetic Polymers

- Natural rubber = Isoprene polymer; Neoprene = Chloroprene polymer. Teflon aur Dacron polymers hain, aur Polythene Ethylene ka polymer hai. [Q28]
- Teflon ka chemical naam PTFE (Polytetrafluoroethylene) hai. Ye non-stick cookware coating mein widely use hota hai. [Q29–Q31]
- PTFE ek solid fluorocarbon / synthetic fluoropolymer hai; iska monomer Tetrafluoroethylene hota hai. "Teflon" PTFE ka common trade name hai. [Q32–Q34]
- Caprolactam khud polymer nahi, balki Nylon production ka monomer/raw material hai. Nylon, Teflon aur Polystyrene polymers hain. [Q35]

6. Fibroin, PFAS & BPA

- Fibroin natural silk protein hai, synthetic nahi. Lexan, Neoprene aur Teflon synthetic polymeric materials hain. [Q36]
- PFAS (per- and polyfluoroalkyl substances) consumer products/food packaging/non-stick applications se associated synthetic chemicals ka large group hai. Strong C–F bonds ke karan ye environment mein persistent rehte hain aur bioaccumulation concern create kar sakte hain. [Q37]
- Bisphenol A (BPA) primarily Polycarbonate plastics aur Epoxy resins ki manufacture mein key chemical hai; food-packaging materials, water bottles aur can coatings ke context mein poocha jata hai. [Q38–Q39]

7. Bakelite — Phenol–Formaldehyde Polymer

- Bakelite ek thermosetting Phenol–Formaldehyde resin hai. Ye Phenol aur Formaldehyde ki condensation reaction se banta hai. [Q40–Q41]
- Bakelite ko Leo Baekeland ne early 20th century mein develop kiya; exam recall ke liye "Phenol + Formaldehyde → Bakelite" direct pair yaad rakho. [Q40–Q41]

✓ Source-question mapping preserved: Q1–Q41